

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2021-22

Paper: GEN-202 — Programming In Python

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt all the Parts. The figures in the right margin indicate the marks.***PART — A***Note: Multiple Choice Type Questions and Short Questions. Each question carries 2 marks.***[10 × 2 = 20]****Question 1.** What will be the output of the following code snippet?

```
a = 3
b = 1
print(a, b)
a, b = b, a
print(a, b)
```

- (a) 3 1 1 3
- (b) 3 1 3 1
- (c) 1 3 1 1
- (d) 1 3 3 1

Question 2. What will be the result of the following command in Python?

```
round(0.4) - round(-0.5)
```

- (a) 0
- (b) 1
- (c) 2
- (d) -1

Question 3. What is the output of the following program in Python?

```
print(0.2 + 0.4 == 0.6)
```

- (a) True
- (b) False
- (c) Error
- (d) Depends on the machine

Question 4. Which type of programming does Python support?

- (a) Object-oriented programming
- (b) Structured programming
- (c) Functional programming
- (d) All of the mentioned

Question 5. All keywords in Python are in:

- (a) Capitalized
- (b) Lower case
- (c) UPPER CASE
- (d) None of the mentioned

Question 6. What will be the output of the following Python code?

```
i = 1
while True:
    if i % 3 == 0:
        break
    print(i)
    i += 1
```

- (a) 1 2 3
- (b) Error
- (c) 1 2
- (d) None of the mentioned

Question 7. What will be the output of the following Python function?

```
min(max(False, -3, -4), 2, 7)
```

- (a) 4
- (b) -3
- (c) 2
- (d) False

Question 8. What arithmetic operators cannot be used with strings in Python?

- (a) *
- (b) -
- (c) +
- (d) All of the mentioned

Question 9. What are the two main types of functions in Python?

- (a) System function and Custom function

- (b) Built-in function & User-defined function
- (c) User function
- (d) None of the mentioned

Question 10. What is the output of the following expression?

```
import math
print(math.pow(3, 2))
```

- (a) 9.0
- (b) None
- (c) 9
- (d) None of the mentioned

PART — B

Note: Internal Choice — Either or Type Questions. Each question carries 5 marks. Attempt any four. **[5 × 4 = 20]**

Question 11. Write a Python program to accept three integer numbers in the same line and find their sum.

OR

Write a Python program to find the sum of two numbers using command line arguments.

Question 12. Write a Python program to display even numbers between m and n .

OR

Write a Python program that displays stars in right-angled triangular form using nested loops.

Question 13. Write a Python program to retrieve the elements from a 2D array and display them using a `for` loop.

OR

Write a function that returns the result of addition, subtraction, multiplication, and division.

Question 14. Write a Python program to calculate factorial values using recursion.

OR

Write a Python program to read all the strings from a text file and display them.

Question 15. Write a Python program to check whether a given character is a vowel or consonant.

OR

Write a Python program to convert Fahrenheit into Celsius.

PART — C

Note: Open choice — Attempt any three out of four questions. Each question carries 10 marks. **[3 × 10 = 30]**

Question 16. Write a Python program to find the L.C.M. of two numbers.

Question 17. How do you create a dictionary and explain its different operations? Elaborate how to iterate over a dictionary.

Question 18. Consider a **Rectangle** class. Write Python code to check if the area of the first rectangle is greater than the second by overloading the > operator.

Question 19. Discuss the technical strengths of Python. What is the difference between a list and a tuple in Python?